

Breast cancer

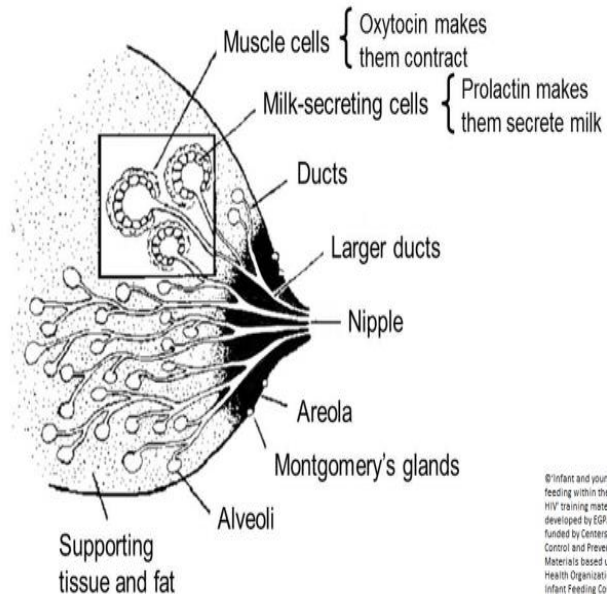
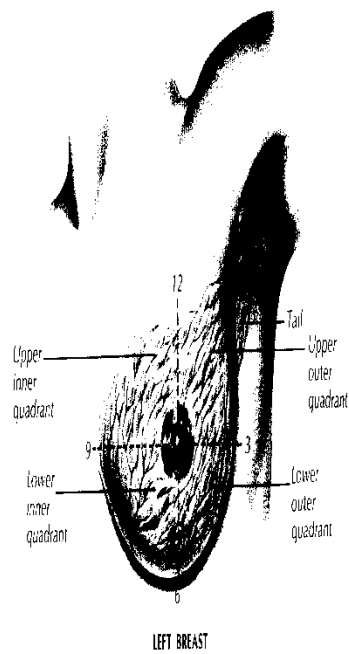
Lecture outline:

- **Definition**
- **Sign and symptoms**
- **Method of diagnosis**
- **Nursing care and rehabilitation**
- **lymphedema**: chronic swelling of an extremity due to interrupted lymphatic circulation, typically from an axillary lymph node dissection
- **BRCA1 and BRCA2**: genes on chromosome 17 that, when damaged or mutated, increase a person's risk for breast or ovarian cancer compared with people without the mutation
- **breast conservation treatment**: surgery to remove a breast tumor and a margin of tissue around the tumor without removing any other part
- **benign proliferative breast disease**: various types of atypical, yet noncancerous, breast tissue that increase the risk of breast cancer
- **mastectomy**: removal of the breast tissue and nipple–areola complex
- **modified radical mastectomy**: removal of the breast tissue, nipple–areola complex, and a portion of the axillary lymph nodes

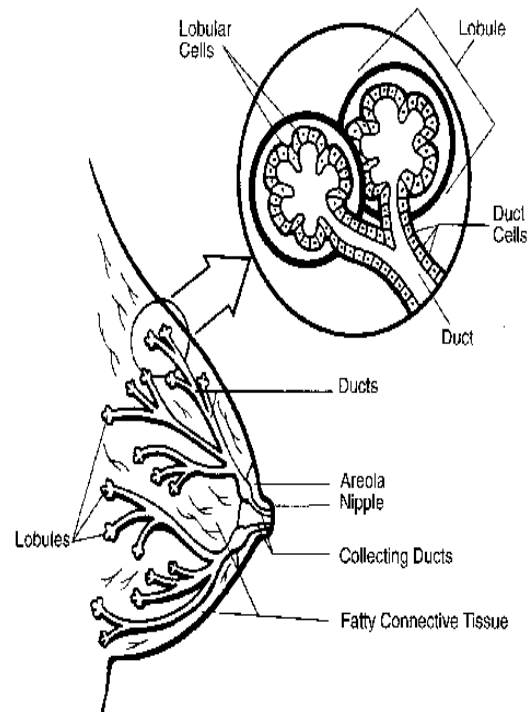
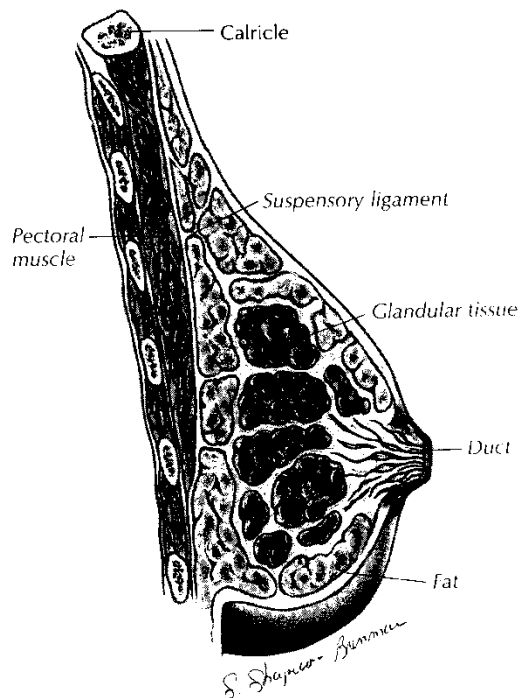
What Is Breast Cancer?

- **Cells in the breast grow out of control and form a lump**
 - **Lumps can either be benign or malignant**
 - **Malignant growths are called breast cancer**

Anatomy of the breast



©Infant and young feeding within the HIV training material developed by EGFH funded by Centers for Disease Control and Prevention. Materials based on Health Organization Infant Feeding Guid.



Normal breast physiology and anatomy

- ▶ Symmetry and balance
- ▶ Size
 - weight
 - menstrual cycle
 - pregnancy and lactation
- ▶ Texture
- ▶ Shape
 - Age

Breast Cancer in USA

One out of *eight* American women will be diagnosed with breast cancer

FIGURES AND FACTS

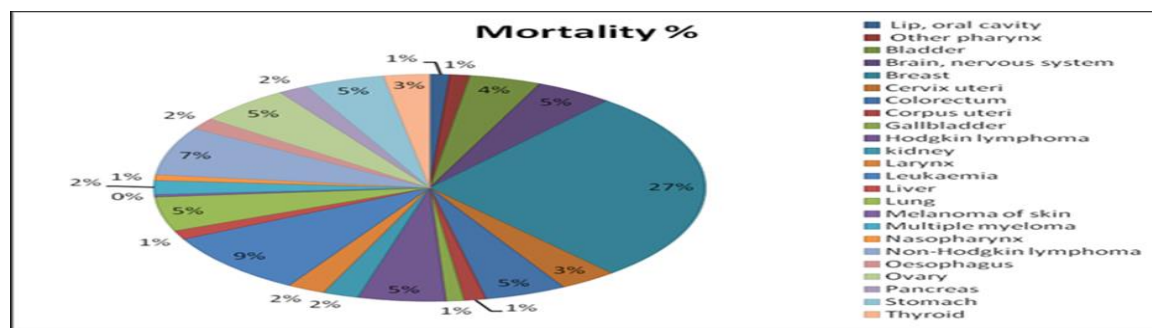
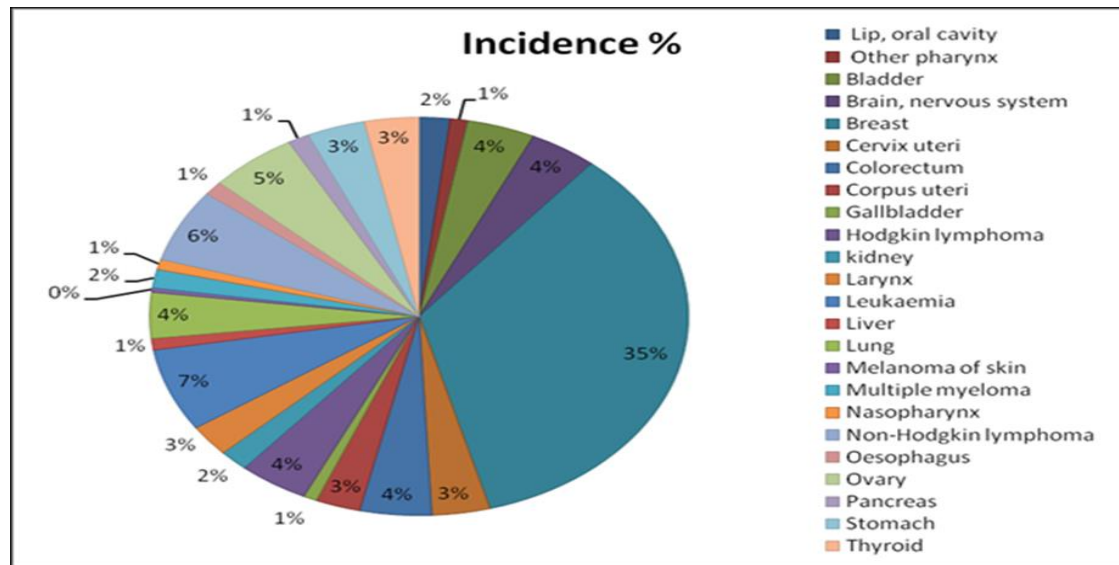
Breast cancer is the most common type of cancer among women in the US and Iraq

Breast Cancer in Iraq

Where it formed the incidence of breast cancer 35% of all cancers in 2008, according to figures the International Agency for Research on Cancer (IARC)
Meanwhile, the mortality rate of 25%

To begin the overview of breast cancer I'd like to present you with some statistics, which unfortunately are still quite grim despite advances in diagnosis and treatment of breast cancer

Breast Cancer in Iraq



Risk Factors for Breast Cancer

- Female (1% male)
- Aging
- Relative (mother or sister)
- Menstrual history
 - early on set
 - late menopause
- Child birth
 - After the age of 30
- Exogenous Estrogen

- ▶ Hormonal replacement therapy(HRT)
 - 30% increased risk with long term use
- ▶ Oral Contraceptives(OC)
 - risk slight
 - risk returns to normal once the use of OC's has been discontinued
- ▶ There is an increased risk of breast cancer associated with the use of exogenous estrogen, such as in hormone replacement therapy (HRT).

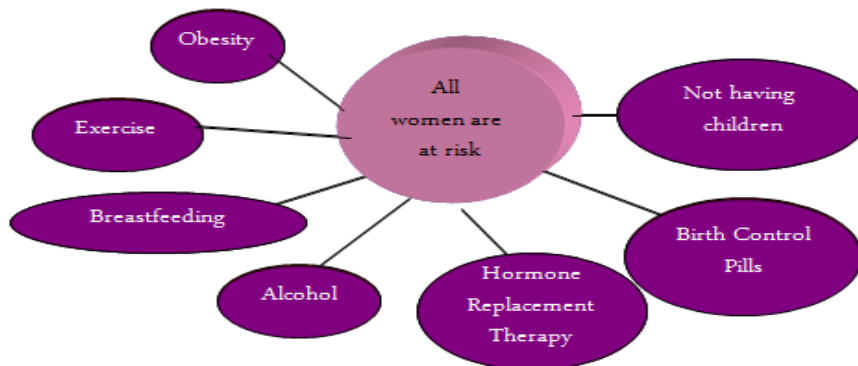
Risk Factors for Breast Cancer

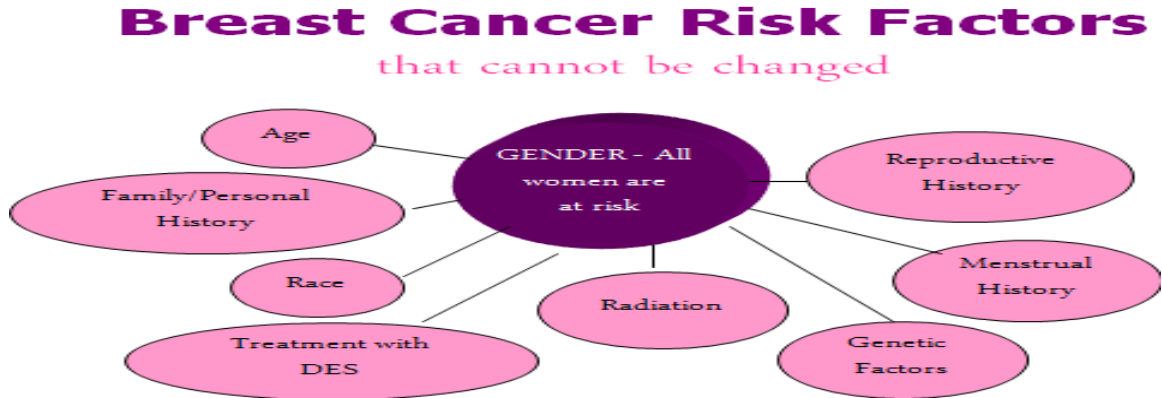
- Radiation exposure
- Breast disease
 - Atypical Hyperplasia
 - Intraductal carcinoma in situ
 - Intralobular carcinoma in situ
- Obesity
- Diet
 - Fat
 - Alcohol

Genetics

- Mutations of Breast cancer genes(BRCA 1 and BRCA 2)
- طفرات جينات سرطان الثدي

Breast Cancer Risk Factors *that can be controlled*





Staging of Breast Cancer

- The American Joint Committee on Cancer (AJCC) has designated staging by TNM
- T= tumor size
- N = lymph node involvement
- M = metastasis

Stages of Tumors

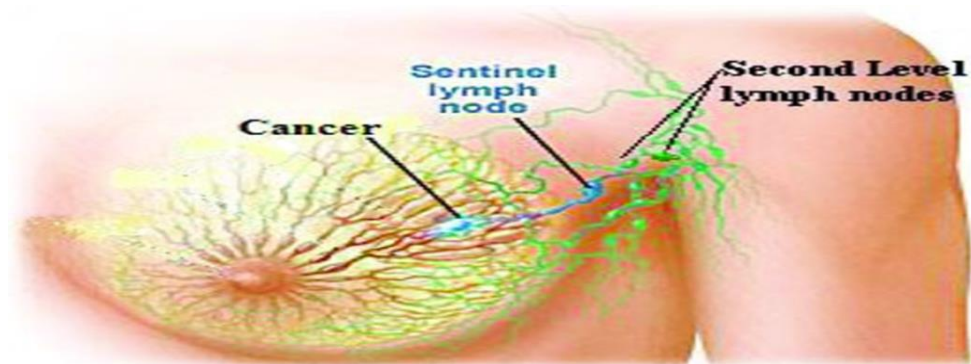
Stage 0 -- carcinoma in situ

„ Stage I – tumor < 2 cm, no nodes

„ Stage II – tumor 2 to 5 cm, +/- nodes

„ Stage III – locally advanced disease, fixed or matted lymph nodes and variable tumor size

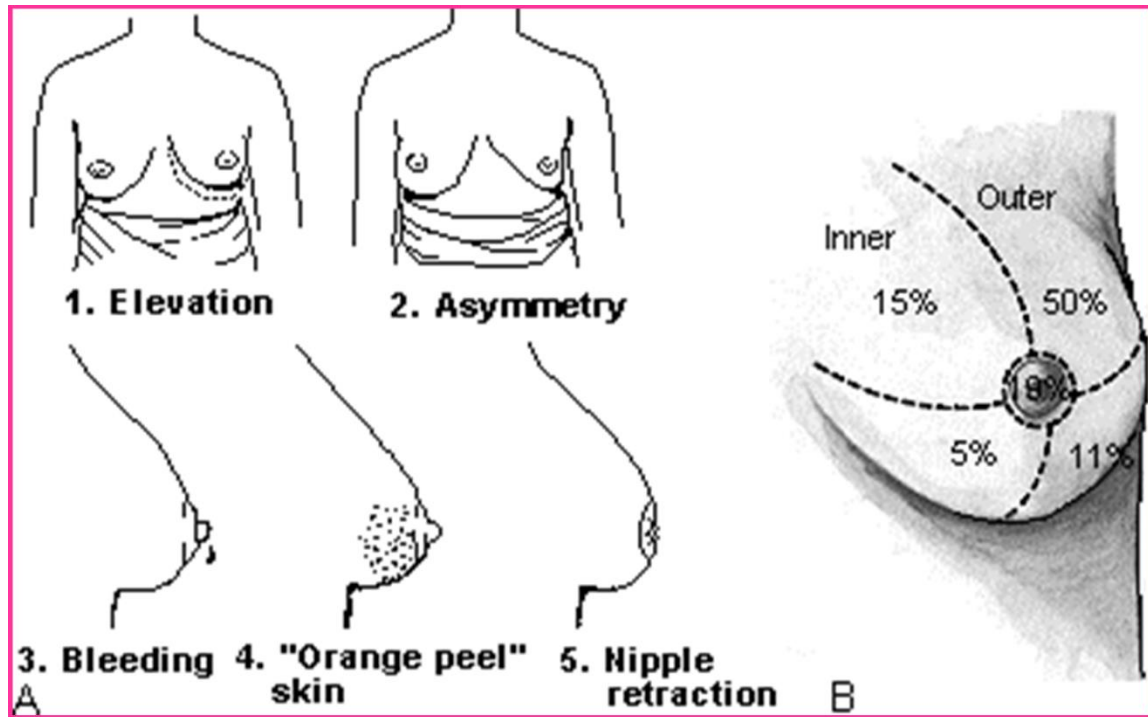
„ Stage IV – distant metastases (bone, liver, lung, brain)



Types of breast cancer

- In situ
 - Intraductal (DCIS)
 - Intralobular (LCIS)
- Invasive
 - Infiltrating ductal carcinoma
 - Tubular carcinoma
 - Medullary carcinoma
 - Mucinous carcinoma
- Non-invasive Ca – ductal in situ or lobular in situ, abnormal mammogram

Symptoms and Screening:



Abnormal signs and symptoms

- Puckeringالتجاعيد
- Dimplingمنكمش
- Retraction
- Nipple discharge
- Thickening of skin or lump or "knot"

- Retracted nipple
- Change in breast size
- Pain or tenderness
- Redness
- Change in nipple position
- Scaling around nipples
- Sore on breast that does not heal

BENIGN BREAST DISORDERS FIBROADENOMA

- Most common cause of breast masses, especially in teens & young women (to early 30's)
- Often upper, outer quadrant
- Solid, slowly enlarging, benign mass, unattached to surrounding breast tissue
- Usually round, firm, easily movable, nontender, clearly distinct from surrounding tissue
- Enlarges slowly



Early detection is the best protection

Methods of Detection:

- Clinical exam by MD or nurse
- Mammography
- Monthly breast self-exam (BSE)
- Fine Needle Aspiration Cytology
- Clinical examination
- Performed by doctor or trained nurse practitioner
- Annually for women over 40
- At least every 3 years for women between 20 and 40
- More frequent examination for high risk patients

Mammography

- Average-size lump found by woman practicing occasional breast self-exam (BSE)
- Average-size lump found by woman practicing regular breast self-exam (BSE)
- Average-size lump found by first mammogram
- Average-size lump found by getting regular mammograms

Mammography

- ▶ X-ray of the breast
- ▶ Has been shown to save lives in patients 50-69
- ▶ Data mixed on usefulness for patients 40-49
- ▶ Normal mammogram does not rule out possibility of cancer completely

Breast Self-Examination

- Opportunity for woman to become familiar with her breasts
 - Monthly exam of the breasts and underarm area
 - May discover any changes early
 - Begin at age 20, continue monthly
- When to do BSE
- ▶ Menstruating women- 5 to 7 days after the beginning of their period
 - ▶ Menopausal women - same date each month
 - ▶ Pregnant women – same date each month
 - ▶ Takes about 20 minutes
 - ▶ Perform BSE at least once a month
 - ▶ Examine all breast tissue

Breast self-examination there is two parts to BSE

- Looking
- Feeling
- exam each breast separately
- use pads of fingers not the tips
- exam the armpits

American Cancer Society Screening Recommendations

- n Annual mammograms, starting at age 40

n Clinical breast exams

- every year starting at age 40
- every 3 years for women age 20-39

n Self-breast exams monthly, starting at age 20

Periodic examinations of the breast: -

Screening/ Age	20-29 years	30-39 years	40-52 years	above 52 years
Breast self-examination	monthly	monthly	monthly	monthly
Clinical breast examination	once every (1-3) years	annually	annually	annually
X ray (mammogram)	----- --	----- ----	All times (1-2) years	All times (1-2) years

Treatment options:

- Size of the tumor
- Invasive or in situ
- Lymph node status
- Whether it has spread other parts of the body

Treatment:

- Surgery mastectomy, lumpectomy
- Radiation therapy
- Chemotherapy
- Hormonal therapy

Breast Cancer Risk Reduction

- Avoid postmenopausal obesity
- Keep alcohol consumption low
- Regular exercise
- Keep an adequate Vitamin D level
- Avoid long term hormone use

Health Professionals Associated With Cancer Care

Medical oncologist (on-KAHL-uh-jist): a doctor who specializes in diagnosing and treating cancer with chemotherapy and other drugs. A medical oncologist is different from a *surgical oncologist*, who mostly treats cancer with surgery.

Palliative (PAL-ee-uh-tiv) **care team**: typically includes a palliative care doctor who leads the team and works with a nurse, social worker, patient navigator, and maybe a person with a spiritual role such as a chaplain or a priest, to treat symptoms but not necessarily the cause of the symptoms.

Radiation oncologist (RAY-dee-A-shun on-KAHL-uh-jist): a doctor who specializes in using radiation to treat cancer.

Radiologist (RAY-dee-AH-luh-jist): a doctor with special training in diagnosing diseases by interpreting (reading) x-rays and other types of imaging studies that make pictures of the inside of the body.

Nursing Care Plan for a woman with Breast Cancer

- Risk for infection, related to surgical incision
- Ineffective tissue perfusion, related to edema
- Acute pain, related to surgery
- Disturbed body image, related to loss of breast
- Decisional conflict about treatment, related to concerns about risks and benefits
- Interrupted family processes, related to effect of surgery and therapy on family roles and relationships
- Fear, related to disease process/prognosis

Nursing Interventions

1. Monitor for adverse effects of radiation therapy such as fatigue, sore throat, dry cough, nausea, anorexia.
2. Monitor for adverse effects of chemotherapy; bone marrow suppression, nausea and vomiting, alopecia, weight gain or loss, fatigue, stomatitis, anxiety, and depression.
3. Realize that a diagnosis of breast cancer is a devastating emotional shock to the woman. Provide psychological support to the patient throughout the diagnostic and treatment process.
4. Involve the patient in planning and treatment.
5. Describe surgical procedures to alleviate fear.
6. Prepare the patient for the effects of chemotherapy, and plan ahead for alopecia, fatigue.
7. Administer antiemetics prophylactically, as directed, for patients receiving chemotherapy.
8. Administer I.V. fluids and hyperalimentation as indicated.
9. Help patient identify and use support persons or family or community.
10. Suggest to the patient the psychological interventions may be necessary for anxiety, depression, or sexual problems.
11. Teach all women the recommended cancer-screening procedures

Documentation Guidelines

- Response to surgical interventions: Condition of dressing and wound, stability of vital signs, recovery from anaesthesia
- Presence of complications: Pain,,edema,,infection, seroma, and ,limited ROM
- Knowledge of and intent to comply with adjuvant therapies
- Reaction to cancer and body changes
- Knowledge of and intent to comply with incision care, postoperative exercises, arm precautions, follow-up care, and early detection methods for recurrence

Discharge and Home Healthcare Guidelines (PATIENT TEACHING)

- Teach precautions to prevent lymphedema after node dissection
- Request no blood pressure or blood samples from affected arm.
- Do not carry packages, handbags, or luggage with the affected arm; avoid elastic cuffs.

- Protect the hand and arm from burns, sticks, and cuts by wearing gloves to do gardening and housework ,using a thimble to sew, applying sunscreen and insect repellent when out-of-doors. Report swelling, pain, or heat in the affected arm immediately. Put the arm above the head and pump the fist frequently throughout the day.

Diagnosis

preoperative nursing diagnoses may include the following:

- Deficient knowledge about breast cancer and treatment options
- Anxiety related to cancer diagnosis
- Fear related to specific treatments, body image changes, or possible death
- Risk for ineffective coping (individual or family) related to the diagnosis of breast cancer and related treatment options
- Decisional conflict related to treatment options

postoperative nursing diagnoses may include the following:

- Acute pain and discomfort associated with surgical procedure
- Risk for impaired peripheral neurovascular function associated with nerve irritation in affected arm, breast, or chest wall
- Disturbed body image associated with loss or alteration of the breast
- Risk for difficulty with coping associated with the diagnosis of cancer and surgical treatment
- Impaired ability to perform hygiene, impaired ability to dress, impaired self feeding, and impaired self toileting associated with partial immobility of upper extremity on operative side
- Lack of knowledge: drain management after breast surgery, arm exercises to regain
- mobility of affected extremity, hand and arm care after axillary lymph node dissection surgery

Potential complications may include the following:

- Lymphedema
- Hematoma/Seroma formation
- Infection
- Changes in sexual function